



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment – 04

Crypt-Arithmetic Problem

Aim

To solve the Crypt-Arithmetic problem using the Backtracking algorithm.

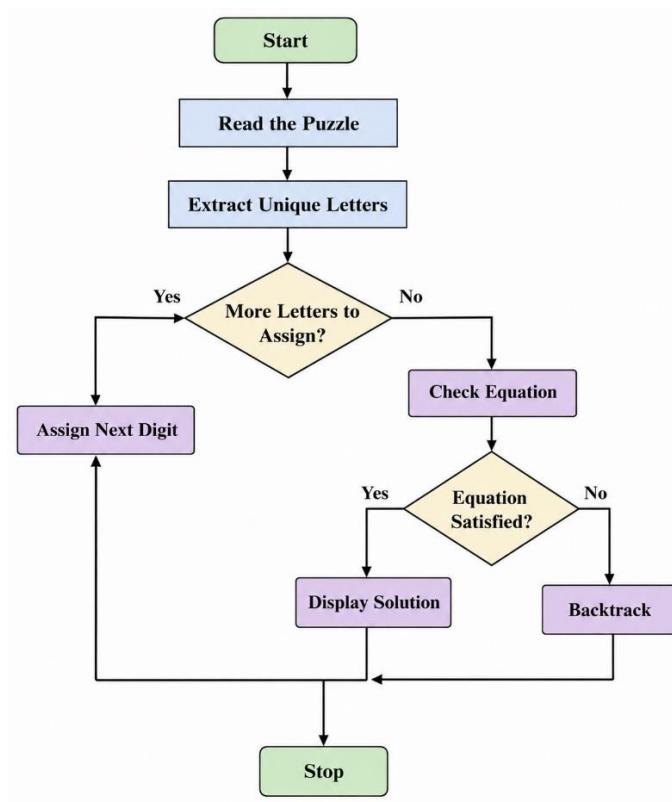
Objective

To find digit assignments to letters in an arithmetic puzzle such that the mathematical equation is satisfied.

Algorithm

1. Read the Crypt-Arithmetic puzzle.
2. Extract all unique letters from the puzzle.
3. Check whether the number of unique letters is less than or equal to 10.
4. Assign digits (0–9) to the letters without repetition.
5. Ensure that the leading letter of each word is not assigned the digit 0.
6. Evaluate the arithmetic expression using the assigned digits.
7. If the equation is satisfied, display the solution.
8. Otherwise, backtrack and try another digit assignment.
9. Repeat until a valid solution is found or all possibilities are exhausted.

FlowChart



Code:

```
from itertools import permutations
```

```
letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
```

```
for p in permutations(range(10), len(letters)):
```

```
    d = dict(zip(letters, p))
```

```
    if d['S'] == 0 or d['M'] == 0:
```

```
        continue
```

```
    send = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
```

```
    more = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
```

```
    money = (10000*d['M'] + 1000*d['O'] +
```

```
            100*d['N'] + 10*d['E'] + d['Y'])
```

```
if send + more == money:

    print("Solution Found:")

    print(d)

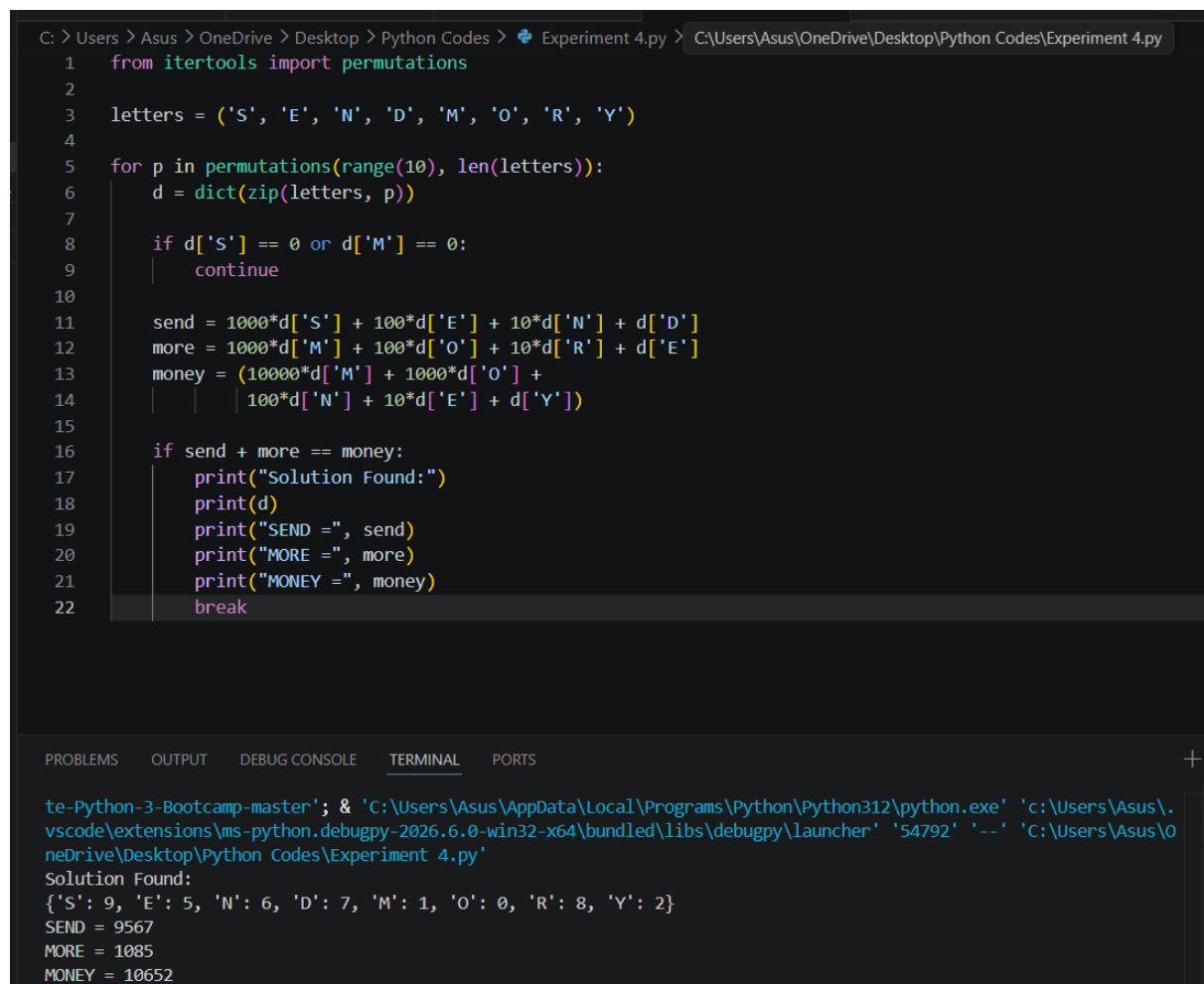
    print("SEND =", send)

    print("MORE =", more)

    print("MONEY =", money)

    break
```

Output



The screenshot shows a VS Code editor with a Python script named 'Experiment 4.py' open. The script uses the 'permutations' function from 'itertools' to find a solution to a cryptarithm. The terminal output shows the execution path and the final solution found.

```
C: > Users > Asus > OneDrive > Desktop > Python Codes > Experiment 4.py > C:\Users\Asus\OneDrive\Desktop\Python Codes\Experiment 4.py
1  from itertools import permutations
2
3  letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
4
5  for p in permutations(range(10), len(letters)):
6      d = dict(zip(letters, p))
7
8      if d['S'] == 0 or d['M'] == 0:
9          continue
10
11     send = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
12     more = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
13     money = (10000*d['M'] + 1000*d['O'] +
14             100*d['N'] + 10*d['E'] + d['Y'])
15
16     if send + more == money:
17         print("Solution Found:")
18         print(d)
19         print("SEND =", send)
20         print("MORE =", more)
21         print("MONEY =", money)
22         break
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
te-Python-3-Bootcamp-master'; & 'C:\Users\Asus\AppData\Local\Programs\Python\Python312\python.exe' 'c:\Users\Asus\
vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '54792' '--' 'C:\Users\Asus\O
neDrive\Desktop\Python Codes\Experiment 4.py'
Solution Found:
{'S': 9, 'E': 5, 'N': 6, 'D': 7, 'M': 1, 'O': 0, 'R': 8, 'Y': 2}
SEND = 9567
MORE = 1085
MONEY = 10652
```

Result

The Crypt-Arithmetic problem was solved successfully using the Backtracking algorithm.

Conclusion

The program successfully assigns unique digits to letters and verifies the arithmetic equation. Backtracking ensures that all possible assignments are explored until a valid solution is found.